Laboratory 07: Logistic Regression
Project results and model evaluation

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May 21, 2026

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Chapter 1

Standard logistic regression analysis

In this section, we analyze the performance of a standard (non-prior-weighted) Logistic Regression model on the project dataset. The goal is to evaluate the impact of the regularization parameter λ on both the discriminative capability (minDCF) and the calibration (actDCF) of the model for a target application characterized by an effective prior $\tilde{\pi} = \pi_T = 0.1$.

1.1 Experimental results

The model was trained using 13 values of λ logarithmically spaced between 10^{-4} and 10^{2}. For each value, we computed the Minimum DCF (minDCF) and Actual DCF (actDCF). To evaluate the actual Bayes decisions, we converted the raw logistic regression scores into valid Log-Likelihood Ratios (LLRs) by subtracting the empirical prior log-odds of the training set.

The numerical results are summarized in Table 1.1 and the visual trend is plotted in Figure 1.1.

Table 1.1: Standard Logistic Regression Performance ($\pi_T = 0.1$)

Regularization Coefficient (λ)	**minDCF**	**actDCF**
1.0000×10^{-4}	0.3640	0.4021
3.1623×10^{-4}	0.3650	0.4051
1.0000×10^{-3}	0.3650	0.4130
3.1623×10^{-3}	0.3641	0.4297
1.0000×10^{-2}	**0.3611**	0.4568
3.1623×10^{-2}	0.3621	0.5805
1.0000×10^{-1}	0.3641	0.8522
3.1623×10^{-1}	0.3640	0.9950
1.0000×10^{0}	0.3640	1.0000
3.1623×10^{0}	0.3640	1.0000
1.0000×10^{1}	0.3640	1.0000
3.1623×10^{1}	0.3640	1.0000
1.0000×10^{2}	0.3640	1.0000

The optimal discriminative capability is reached at $\lambda = 10^{-2}$, yielding a minDCF of **0.3611**. However, at this specific operational point, the actual DCF is noticeably higher at 0.4568, indicating a baseline calibration gap.

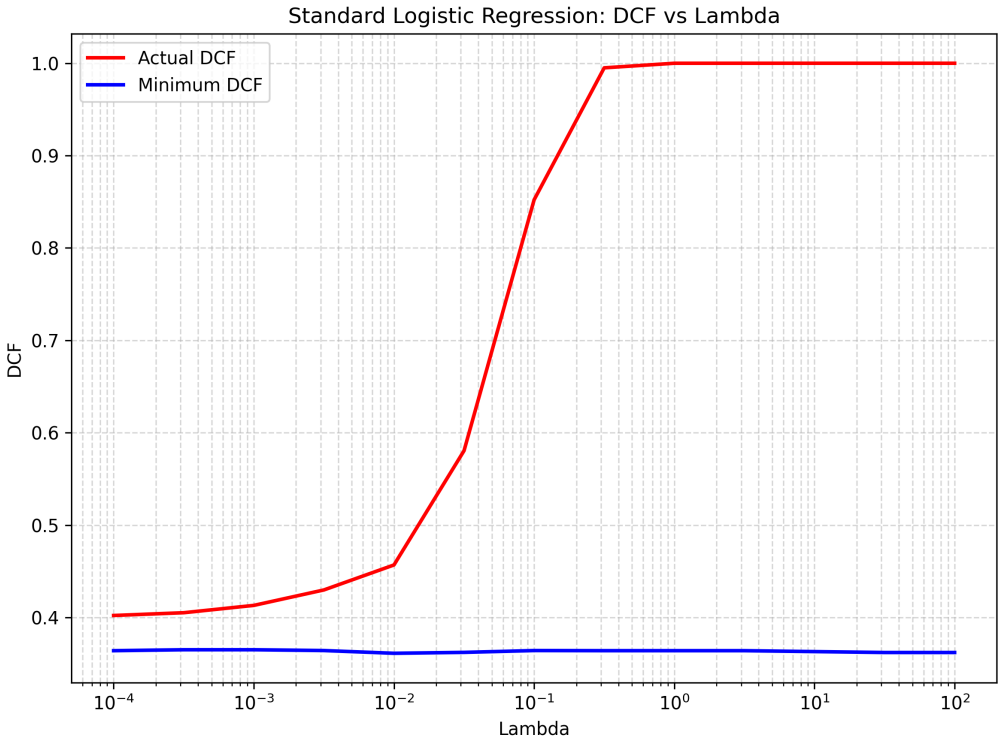


Figure 1.1: Actual DCF and Minimum DCF as a function of the regularization parameter λ for the Standard Logistic Regression model ($\pi_T = 0.1$).

1.2 Analysis and considerations

Does the regularization coefficient affect the two metrics differently?

Yes, as vividly demonstrated in Figure 1.1, the parameter λ affects the two metrics in completely different ways. The minimum DCF (blue line) remains remarkably flat across the entire parameter sweep, hovering consistently around 0.364. There are no significant differences in terms of the model's potential discriminatory power as regularization increases.

Conversely, there is a severe deterioration in terms of calibration (red line). The model maintains a reasonable, albeit imperfect, calibration for low values of $\lambda \leq 10^{-3}$. However, as regularization increases beyond 10^{-2}, the actDCF skyrockets rapidly, eventually maxing out at a worst-case value of 1.0000 for $\lambda \geq 1$. This clearly shows that λ has a much stronger, destructive impact on the reliability of the scores (actDCF) than on their relative ranking ability (minDCF).

How does the regularization coefficient affect the two metrics?

The regularization coefficient λ affects the two metrics in fundamentally different ways due to its mathematical interaction with the Logistic Regression objective function:

- **Effect on minDCF:** The impact is negligible. The minimum DCF evaluates the ability of the classifier to correctly rank samples from highest to lowest probability of belonging to the target class. Because our training dataset has plenty of samples relative to the small number of features (6 dimensions), the model is not prone to overfitting in this linear configuration. Therefore, imposing an L_2 penalty does not change the optimal orientation of the separating hyperplane. The relative ordering of the samples is preserved, which keeps the minDCF flat.

- **Effect on actDCF:** Regularization has a severe negative impact on actDCF. Theoretically,

Logistic Regression minimizes the cross-entropy to produce outputs that can be mapped to Log-Likelihood Ratios. However, the regularization term $\frac{\lambda}{2}||w||^2$ penalizes the magnitude of the weight vector w. As λ grows large ($\lambda \to \infty$), the optimization forces the weights to shrink towards zero ($w \to 0$).

When the weights are extremely small, the raw scores $s(x) = w^T x + b$ become dominated by the bias term b, losing their variance and pulling the computed LLRs toward zero. For a highly unbalanced target application like $\pi_T = 0.1$, the optimal theoretical Bayes threshold is high ($t = -\log \frac{0.1}{0.9} \approx 2.19$). If the L2 penalty artificially shrinks all test scores well below this threshold, the model will constantly predict the negative class for *every single sample*, leading to a False Negative Rate of 100% and an actDCF of 1.0000. This completely destroys the calibration.

Chapter 2

Reduced dataset analysis

In this section, we analyze the behavior of the Logistic Regression model when training data is severely limited. By keeping only 1 out of every 50 samples from the original training set, we drastically reduce the number of observations available for parameter estimation. This setup allows us to highlight the vital role of explicit regularization as a tool to prevent overfitting in low-data regimes.

2.1 Experimental results

The model was trained on the reduced dataset $(DTR_{reduced}, LTR_{reduced})$ using the exact same sweep of 13 λ values as the standard analysis. The validation was still performed on the full validation set to ensure a reliable and comparable estimate of the generalization error for the target application $\pi_T = 0.1$.

The numerical results are summarized in Table 2.1 and the visual trends are plotted in Figure 2.1.

Table 2.1: Reduced dataset Logistic Regression results ($\pi_T = 0.1$)

Regularization Coefficient (λ)	**minDCF**	**actDCF**
1.0000×10^{-4}	0.4466	0.9780
3.1623×10^{-4}	0.4487	0.9780
1.0000×10^{-3}	0.4487	0.7466
3.1623×10^{-3}	0.4431	**0.4652**
1.0000×10^{-2}	0.4407	**0.4652**
3.1623×10^{-2}	0.4407	**0.4652**
1.0000×10^{-1}	0.3988	0.7164
3.1623×10^{-1}	0.3874	0.9856
1.0000×10^{0}	0.3874	1.0000
3.1623×10^{0}	0.3874	1.0000
1.0000×10^{1}	**0.3783**	1.0000
3.1623×10^{1}	0.3807	1.0000
1.0000×10^{2}	0.3920	1.0000

The optimal discriminative capability is reached at a highly regularized $\lambda = 10^1$, yielding a minDCF of **0.3783**. Conversely, the best calibration occurs much earlier, between $\lambda = 3.16 \times 10^{-3}$ and 3.16×10^{-2}, where the actDCF drops to a minimum of **0.4652**.

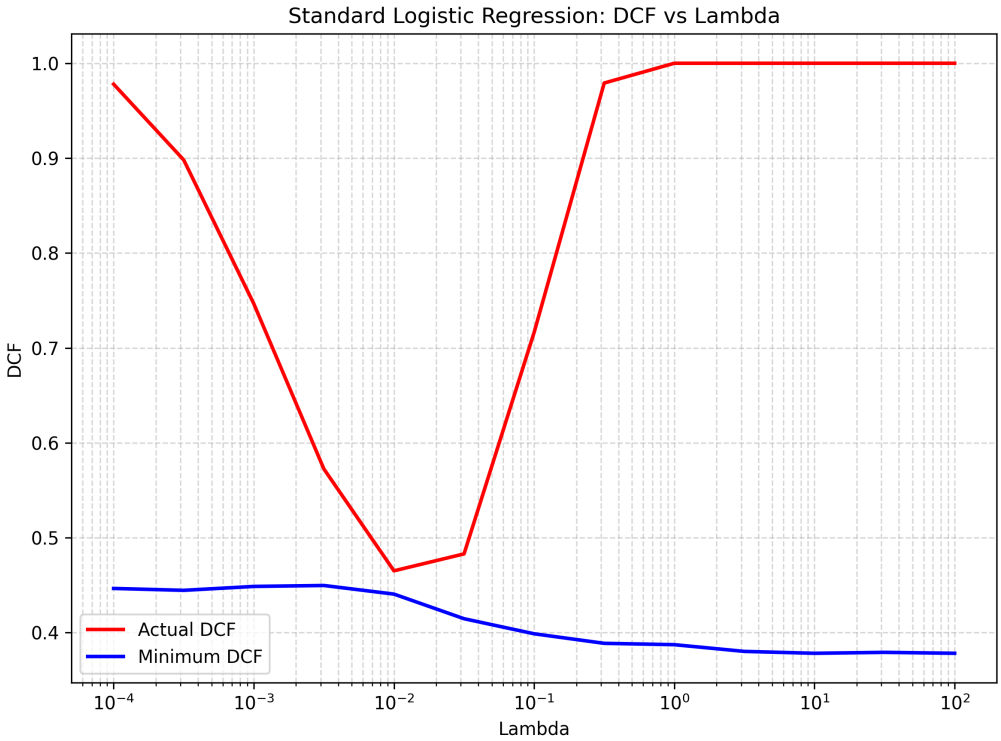


Figure 2.1: Actual DCF and Minimum DCF as a function of λ for the Standard Logistic Regression model trained on the reduced dataset (1/50 samples).

2.2 Analysis and Considerations

By comparing Figure 2.1 with the standard plot from the previous chapter, we can observe entirely different model dynamics dictated by the scarcity of data. We can break this behavior down into three distinct regularization regimes:

- **Low regularization ($\lambda \leq 10^{-3}$): Overfitting.** Unlike the full dataset model where minDCF was flat, here a low λ produces terrible results for both metrics (minDCF $\approx$ 0.448, actDCF $\approx$ 0.978). With very few training samples and virtually no penalty on the weights, the model completely memorizes the noise in the tiny training set. The resulting weights become extremely large and highly tailored to those specific points, destroying the model's ability to generalize to the evaluation set. The model produces wildly overconfident, miscalibrated scores, ruining both its ranking ability and its absolute thresholding.

- **Moderate regularization ($\lambda \approx 10^{-2}$): Calibration recovery.** As we increase λ, we penalize extreme weight values. This forces the model to ignore spurious correlations (noise) and rely only on the most robust, generalizable features. This variance reduction significantly improves the reliability of the scores, bringing the empirical risk closer to the true risk. This results in the best-calibrated scores and the lowest actDCF (0.4652).

- **High regularization ($\lambda \geq 10^{-1}$): Underfitting and calibration loss.** As λ increases further, the penalty term heavily dominates the cross-entropy loss. The weights w are aggressively pushed toward zero, causing the model to become too rigid to capture the underlying data distribution (underfitting). Just as in the standard model, these "shrunken" weights compress the scores $s(x) = w^T x + b$ toward a constant value dictated entirely by the bias. The scores lose their probabilistic meaning, forcing the system to predict the negative class for all samples. Consequently, the actDCF hits the maximum ceiling of 1.0000.

Why does minDCF continue to improve at high λ?

An interesting phenomenon in the reduced dataset is that the minDCF actually reaches its absolute best value (0.3783) at $\lambda = 10^1$, well into the regime where actDCF is completely broken (1.0000).

This happens because minDCF only cares about the *relative sorting* of the samples. In a severely data-starved environment, heavily squashing the weights limits the model to acting essentially like a mean-distance classifier. While the absolute values of these heavily regularized scores are useless for direct thresholding (hence actDCF = 1), their relative ranking is actually more stable and robust than the erratic, overfitted ranking produced by a low λ.

2.3 Theoretical Conclusion

In conclusion, when training data is abundant, the model is naturally regularized by the sheer volume of information, rendering the L2 penalty mostly irrelevant for discriminative power. However, when data is scarce, explicit regularization via λ is strictly mandatory. It is the only mechanism available to control the bias-variance tradeoff, preventing catastrophic overfitting and allowing the model to recover a minDCF (0.3783) that is surprisingly close to the full-dataset baseline (0.3611).

Chapter 3

Prior-weighted logistic regression analysis

In this section, we analyze the performance of the Prior-weighted Logistic Regression model. Unlike the standard model, this version explicitly incorporates the target application prior $\pi_T = 0.1$ into the objective function during the training phase.

3.1 Experimental results

The model was trained on the full project dataset using 13 values of λ logarithmically spaced between 10^{-4} and 10^{2}. To obtain the Log-Likelihood Ratios (LLRs), we subtracted the log-odds of the target prior used during training ($\pi_T = 0.1$) instead of the empirical prior.

The numerical results are summarized in Table 3.1 and plotted in Figure 3.1.

Table 3.1: Prior-weighted logistic regression results ($\pi_T = 0.1$)

Regularization Coefficient (λ)	**minDCF**	**actDCF**
1.0000×10^{-4}	0.3721	0.4071
3.1623×10^{-4}	0.3701	0.4010
1.0000×10^{-3}	0.3699	0.4129
3.1623×10^{-3}	0.3670	0.4327
1.0000×10^{-2}	0.3630	0.4487
3.1623×10^{-2}	0.3640	0.5964
1.0000×10^{-1}	0.3648	0.9147
3.1623×10^{-1}	0.3640	1.0000
1.0000×10^{0}	0.3640	1.0000
3.1623×10^{0}	0.3630	1.0000
1.0000×10^{1}	0.3630	1.0000
3.1623×10^{1}	**0.3620**	1.0000
1.0000×10^{2}	**0.3620**	1.0000

The optimal discriminative capability is reached at high regularization ($\lambda = 3.16 \times 10^{1}$), yielding a minDCF of **0.3620**. The best calibration is observed at $\lambda = 3.16 \times 10^{-4}$, where actDCF hits **0.4010**.

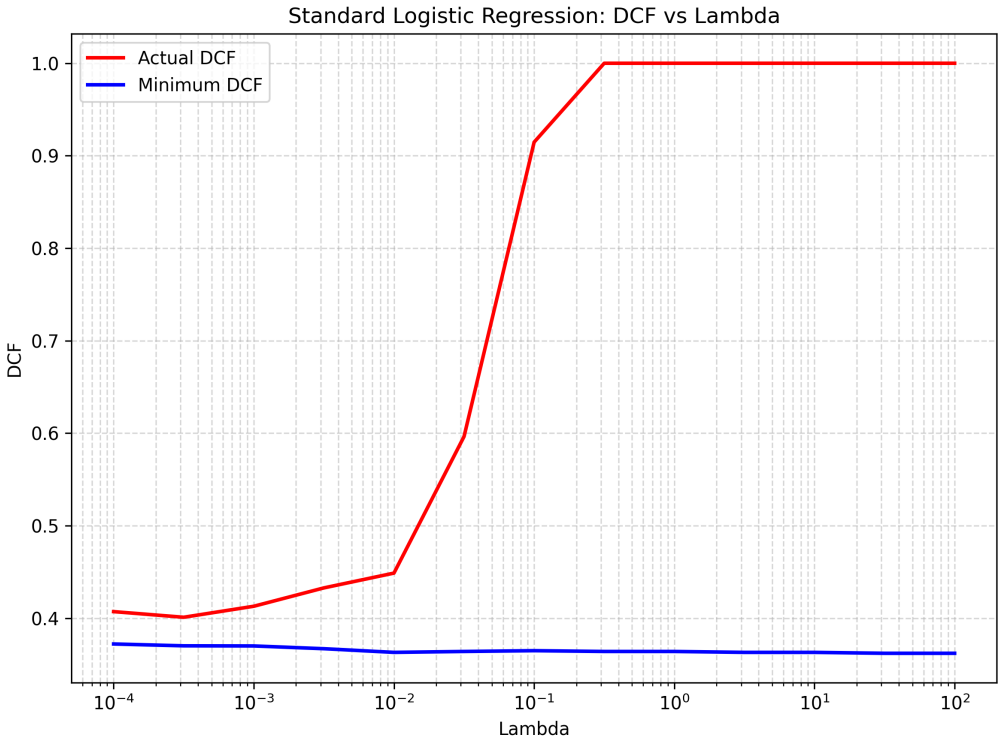


Figure 3.1: Actual DCF and Minimum DCF as a function of λ for the Prior-weighted Logistic Regression model ($\pi_T = 0.1$).

3.2 Analysis and discussion

Are there significant differences for this task?

No, there are no significant differences in terms of performance between the standard and the prior-weighted models for this specific task. Comparing the best minimum DCF obtained by the standard model (0.3611) with the best minimum DCF of the prior-weighted model (0.3620), the values are nearly identical. Both models exhibit the exact same trend: a stable minDCF across the lambda range and a degrading actDCF that rapidly reaches 1.0 as regularization increases past 10^{-1}.

The lack of difference is motivated by the fact that the project dataset is sufficiently large. In high-data regimes, the optimization algorithm has enough information to learn the underlying class distributions and position the separating hyperplane effectively, regardless of whether the minority class errors are explicitly over-weighted in the objective function. The post-hoc threshold adjustment (removing empirical prior log-odds) used in the standard model is mathematically sufficient to recover the optimal Bayes decisions.

Are there advantages using the prior-weighted model for our application?

(Remember that the prior-weighted model requires that we know the target prior when we build the model).

In this specific case, the prior-weighted model does not offer a numerical advantage, but it provides a distinct theoretical one. The primary advantage of the prior-weighted model is its ability to "focus" the training on a specific operating point. By incorporating π_T directly into the loss function, we instruct the optimizer to penalize classification errors on the expected minority class more heavily (in our case, the "Genuine" class since $\pi_T = 0.1$ implies a 90% fake distribution).

Theoretically, the prior-weighted model is most advantageous when the training set is highly imbalanced or completely unrepresentative of the real-world application, allowing the model to adapt its internal weights natively to the deployment scenario.

However, this approach introduces two major practical drawbacks:

- **Requirement of prior knowledge:** We must know the exact target application prior at training time. If the operational requirements change (e.g., if the system administrator later decides to evaluate for a balanced prior of $\pi_T = 0.5$), the prior-weighted model trained specifically for 0.1 will be poorly calibrated for the new threshold and will yield sub-optimal results.
- **Calibration vulnerability:** As seen in the results, even with the "correct" prior explicitly weighted into the training loop, the actDCF still massively diverges from the minDCF at high λ values. This confirms that aggressive L_2 regularization still compresses the scores toward the bias term, leading to a total loss of probabilistic meaning that the prior weighting alone cannot salvage.

Chapter 4

Quadratic logistic regression analysis

In this section, we move beyond linear decision boundaries by applying a quadratic feature expansion to the project dataset. By mapping the original features into a higher-dimensional space that includes all pairwise products of the features ($\phi(x) = [\text{vec}(xx^T), x]^T$), we allow the Logistic Regression model to learn complex, non-linear boundaries in the original feature space.

4.1 Experimental results

The model was trained on the full expanded dataset using a standard (non-prior-weighted) objective function. We evaluated the performance across the same 13 logarithmically spaced λ values for the primary application $\pi_T = 0.1$.

The numerical results are summarized in Table 4.1 and the visual trends are plotted in Figure 4.1.

Table 4.1: Quadratic logistic regression results ($\pi_T = 0.1$)

Regularization Coefficient (λ)	**minDCF**	**actDCF**
1.0000×10^{-4}	0.2602	0.2768
3.1623×10^{-4}	0.2612	**0.2656**
1.0000×10^{-3}	0.2587	0.2765
3.1623×10^{-3}	0.2527	0.2771
1.0000×10^{-2}	0.2487	0.3464
3.1623×10^{-2}	**0.2436**	0.4972
1.0000×10^{-1}	0.2466	0.7520
3.1623×10^{-1}	0.2629	0.9623
1.0000×10^{0}	0.2840	1.0000
3.1623×10^{0}	0.3090	1.0000
1.0000×10^{1}	0.3243	1.0000
3.1623×10^{1}	0.3263	1.0000
1.0000×10^{2}	0.3263	1.0000

The optimal discriminative capability is reached at an intermediate regularization of $\lambda = 3.16 \times 10^{-2}$, yielding an outstanding minDCF of **0.2436**. However, the best calibration occurs at a much lower regularization ($\lambda = 3.16 \times 10^{-4}$), where actDCF drops to an excellent **0.2656**.

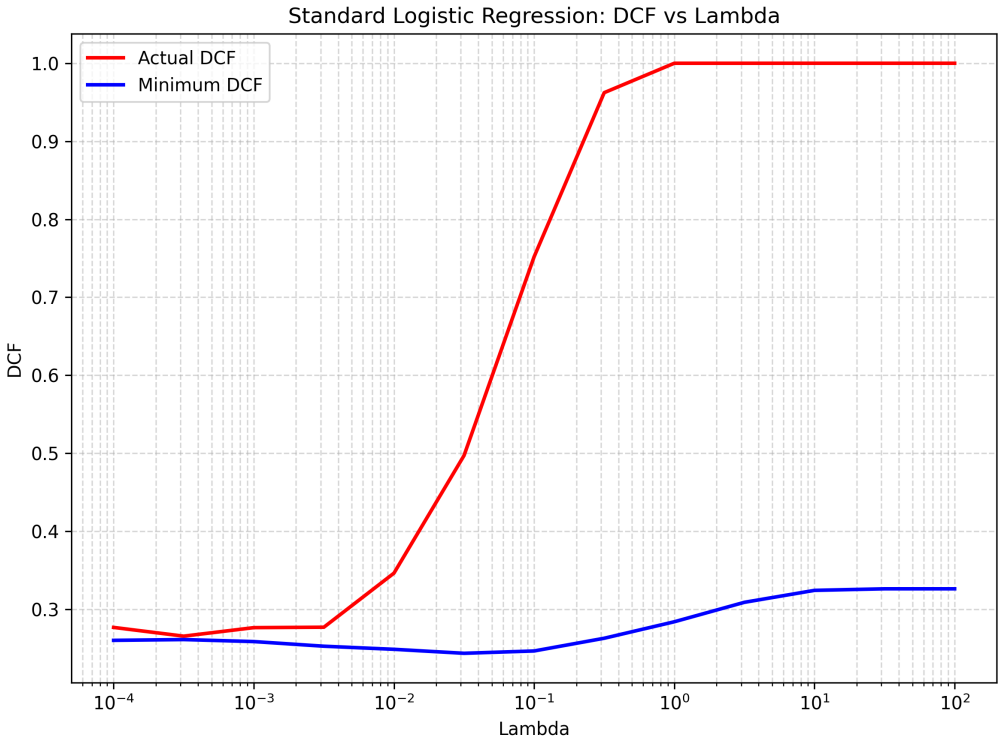


Figure 4.1: Actual DCF and Minimum DCF as a function of λ for the Quadratic Logistic Regression model ($\pi_T = 0.1$).

4.2 Analysis and discussion

What do you observe?

The most striking observation is the massive, systemic improvement in discriminatory power compared to all previously evaluated linear models. The minimum DCF drops precipitously from approximately 0.361 in the linear case down to a best value of 0.2436 in this quadratic case. This mathematically proves that the relationship between the features and the classes is inherently non-linear, and the quadratic expansion successfully maps this complexity.

Furthermore, we observe that the minDCF curve (blue line in Figure 4.1) is no longer flat as it was in the linear full-dataset analysis. Instead, it exhibits a distinct "U-shape," slowly improving as λ increases toward 10^{-2} before deteriorating as the penalty becomes too aggressive.

The actDCF (red line) behaves exceptionally well for small values of $\lambda \leq 10^{-3}$, remaining tightly coupled to the minDCF curve and showing excellent initial calibration (e.g., 0.2656 actDCF vs 0.2612 minDCF). However, it degrades sharply as regularization scales up, hitting the 1.0000 ceiling exactly like the linear models.

In this case is regularization effective?

Yes, in the quadratic case, explicit regularization is highly effective and absolutely necessary for optimizing discriminative power.

In the standard linear analysis on the full dataset, the number of parameters was tiny (6 features + 1 bias = 7 weights). With thousands of training samples, the model was immune to variance, making L_2 regularization largely irrelevant for the minDCF. However, the quadratic expansion drastically increases the dimensionality of the feature space. Computing the outer product xx^T inflates the parameter count substantially, providing the model with a massively expanded capacity.

With this high capacity comes a severe vulnerability to overfitting. We can see regularization counteracting this by observing the minDCF: it improves from 0.2602 at $\lambda = 10^{-4}$ down to the global optimum of 0.2436 at $\lambda = 3.16 \times 10^{-2}$. This proves that penalizing the expanded weight vector prevents the model from mapping the random noise generated by the squared feature terms, locking in a more robust geometric boundary.

How does it affect the two metrics?

The regularization coefficient λ controls the bias-variance tradeoff, affecting the metrics as follows:

- **Effect on minDCF:** For very low λ, the high-capacity model slightly overfits the training set (higher minDCF due to variance). As λ increases toward 3.16×10^{-2}, the minDCF reaches its absolute optimum because the L_2 penalty forces the model to ignore spurious quadratic interactions while preserving the core non-linear class signals. Beyond this point, the model begins to heavily underfit (higher minDCF due to bias), as the massive penalty prevents the weights from drawing any meaningful boundaries.
- **Effect on actDCF:** For low λ, calibration is remarkably accurate because the expanded quadratic model possesses the structural complexity required to accurately model the true log-posterior probabilities without needing severe weight suppression. However, as λ increases beyond the optimal point, the actDCF explodes. Just as in the linear variants, heavy regularization forcefully shrinks the complex weight vector toward zero, artificially compressing the output scores toward the bias term. These compressed scores lose their alignment with the theoretical Bayes threshold ($t \approx 2.19$ for $\pi_T = 0.1$), resulting in a massive increase in the empirical cost.

4.3 Theoretical motivation

The overwhelming superiority of the quadratic Logistic Regression model (0.2436 minDCF) stems from its ability to model covariance differences between classes using a purely discriminative optimization approach.

A standard linear logistic regression is mathematically analogous to a Tied Covariance Gaussian model, in that both are strictly constrained to drawing a flat, linear hyperplane. By passing the outer product xx^T into the model, the quadratic expansion allows the Logistic Regression algorithm to natively compute boundary curvatures, effectively simulating the behavior of a fully unconstrained Multivariate Gaussian (MVG) model.

In fact, the Quadratic LR outperforms our baseline Generative MVG (which scored a 0.2629 minDCF). This happens because the Generative MVG strictly assumes the underlying data follows a perfect Normal distribution—an assumption we know is violated by the heavy skewness of Features 5 and 6 in our dataset. The Quadratic Logistic Regression algorithm learns the non-linear boundaries directly from the error gradients, entirely bypassing the rigid, flawed Gaussian density assumption, thereby extracting the absolute maximum discriminative utility from the feature space.

Chapter 5

Final model comparison and conclusions

In this final section, we consolidate the results obtained across all experimental setups, integrating the generative Gaussian models from the previous laboratory with the discriminative Logistic Regression models developed in this section. The comparison evaluates both the optimal discriminatory potential (minimum DCF) and the empirical operational cost (actual DCF) for the primary application $\pi_T = 0.1$.

5.1 Summary of results

The following table summarizes the best performance achieved by each model type. For Logistic Regression models, the results corresponding to the optimal regularization parameter λ are reported alongside their respective calibration gap.

Table 5.1: Overall model comparison ($\pi_T = 0.1$)

Model Architecture	**Optimal λ**	**minDCF**	**actDCF**	**Calibration Gap**
MVG (Standard)	-	0.2629	0.3051	0.0422
Naive Bayes	-	0.2570	0.3022	0.0452
Tied MVG	-	0.3628	0.4061	0.0432
Standard Logistic Regression	1.00×10^{-2}	0.3611	0.4568	0.0956
Reduced Standard LR (1/50)	1.00×10^{1}	0.3783	1.0000	0.6217
Prior-Weighted Logistic Regression	3.16×10^{1}	0.3620	1.0000	0.6380
Quadratic Logistic Regression	3.16×10^{-2}	**0.2436**	0.4972	0.2536

5.2 Analysis and comparative discussion

Which model(s) achieve(s) the best results?

In terms of pure discriminative power, the absolute best result is achieved by the **Quadratic Logistic Regression** model, which reaches the global minimum DCF of **0.2436**.

Following closely behind are the unconstrained generative models: **Naive Bayes** achieves a highly competitive minDCF of **0.2570**, and the **Standard MVG** follows at **0.2629**. Conversely, all models constrained to linear boundaries (Standard LR, Prior-Weighted LR, and Tied MVG) perform significantly worse, hitting a hard performance ceiling and plateauing at a minDCF around 0.361 to 0.362.

What kind of separation rules or distribution assumptions characterize this / these model(s)?

The top-performing models (Quadratic LR, Naive Bayes, Standard MVG) all share one critical mathematical characteristic: they compute **non-linear (quadratic) decision boundaries**.

- **Quadratic Logistic Regression:** This is a discriminative model that does not make explicit assumptions about the underlying probability density of the features. By expanding the feature space to include quadratic cross-terms, the model optimizes a hyperplane in a higher-dimensional space, which maps back to a curved, quadratic decision boundary in the original feature space.
- **Standard MVG and Naive Bayes:** These are generative models. They explicitly assume that the data within each class follows a Normal distribution. Because they estimate class-specific covariance matrices (allowing the geometric spread of the "Fake" and "Genuine" classes to differ), the resulting Log-Likelihood Ratio forms a quadratic decision boundary.
- **Linear Models (Standard LR, Tied MVG):** These models are strictly limited to linear separating hyperplanes. Tied MVG forces this mechanically by assuming both classes share the exact same covariance matrix. Standard LR achieves this by optimizing a simple linear combination of the raw features without calculating interacting cross-terms.

How are the results related to the characteristics of the dataset features?

By cross-referencing these final metrics with our exploratory analysis from previous laboratories, we can definitively explain the success and failure of these models based on the physical characteristics of the dataset:

1. **Covariance Mismatch (Why Linear Models Fail):** The fact that Quadratic LR (0.2436) and Standard MVG (0.2629) vastly outperform Tied MVG (0.3628) proves that the two classes possess fundamentally different covariance structures. As observed in Laboratory 4, Genuine Feature 1 has a massively different variance compared to Fake Feature 1. A linear model cannot capture this difference in geometric spread, condemning it to a high error rate.
2. **Non-Gaussian Skewness (Why Quadratic LR Wins):** While MVG performs well, the Quadratic Logistic Regression outperforms it. This is directly related to Features 5 and 6. In previous analyses, we proved these features are heavily skewed and do not follow a symmetric Gaussian curve. The generative MVG model suffers because its strict Normal assumption is physically violated by the data. As a discriminative model, the Quadratic LR ignores the density distributions entirely and focuses strictly on optimizing the boundary, making it robust against the skewness of Features 5 and 6 while still exploiting their variance differences.
3. **Feature Independence (Why Naive Bayes Excels):** The Naive Bayes model (0.2570) slightly outperforms the full MVG model (0.2629). This relates back to our Laboratory 5 correlation analysis, where we proved the features are exceptionally weakly correlated (all off-diagonal coefficients hovered near zero). Because the features are already practically independent, the Naive Bayes assumption is highly accurate. By forcing the off-diagonal covariances to zero, the Naive Bayes model acts as a powerful regularizer, eliminating parameter estimation noise and slightly improving upon the full MVG.

5.3 Final considerations on calibration

While Quadratic Logistic Regression provides the absolute best discriminatory potential (lowest minDCF), it suffers from a significant **calibration gap** of 0.2536, yielding an actual operational DCF of 0.4972. This massive loss of calibration is driven by the L_2 penalty term ($\lambda = 3.16 \times 10^{-2}$) compressing the weights and squashing the output scores.

In stark contrast, the generative models prove to be vastly superior in native calibration. The Standard MVG and Naive Bayes models both exhibit tiny calibration gaps of just 0.0422 and 0.0452. Because generative models derive their scores from mathematically rigorous probability densities rather than regularized geometric distances, their outputs naturally behave as true Log-Likelihood Ratios.

This reinforces a fundamental conclusion: while highly complex, regularized discriminative models (like Quadratic LR) can locate superior classification boundaries, their raw scores cannot be reliably thresholded out-of-the-box. Deploying such a model in a real-world security application requires a dedicated, post-processing score calibration stage. Until such calibration is applied, the generative **Naive Bayes** model remains the most reliable, "ready-to-deploy" solution for this dataset.